Multi-Workspace Memory Layer (MWML)

Architecture & Capabilities Overview

What It Is

A locally-hosted, AI-built knowledge consolidation system that ingests years of unstructured creative and strategic work - scattered across multiple AI platforms (ChatGPT, Grok, Claude), local files (markdown, PDFs, Google Docs, spreadsheets), and desktop documents - and transforms it into a searchable, navigable, living knowledge base with full provenance tracking, functioning as the persistent episodic and sensory memory layer of a larger memory system

The system produces an interactive single-page application (HTML) that runs entirely from a local file in a browser. No server, no cloud hosting, no external dependencies. All data stays on the user's machine.

What this layer covers and what it doesn't

The Memory Layer is the persistent episodic and sensory memory of the unified intelligence stack, with declarative concept extraction as a derived semantic layer. It does NOT cover working memory (handled by active LLM context) or procedural memory (handled by agent skills and workflow systems like OmegaClaw). It is one lobe within a larger memory architecture. Other memory layers can be added without disturbing this one.

The Problem It Solves

When a researcher or creator works across multiple AI platforms over months or years, critical ideas, frameworks, decisions, and creative output become fragmented across hundreds of conversations and files. Key problems include:

- **Ideas are unretrievable.** A breakthrough insight from a conversation eight months ago on one platform can't be found. The user knows it exists but can't locate it.
- **Context is lost.** The same concept was discussed in different conversations, each adding nuance - but there's no way to see the full evolution in one place.
- **Conversations mix topics.** A single long AI chat might cover five different subjects. There's no way to extract and organize the relevant parts of each.
- **No cross-platform view.** Work done on ChatGPT, Grok, and Claude lives in three separate, unsearchable silos.

- **No status tracking.** There's no way to know which ideas are well-developed, which are thin, which have contradictions, or which haven't been revisited in months.
- **No provenance.** When an idea is referenced later, there's no way to trace it back to the exact conversation, line, and date where it originated.

Architecture

Multi-Workspace Design

The system is organized around workspaces - independent but linkable knowledge domains. Each workspace functions as its own memory with its own concept registry, status tracking, and visual map. Examples of workspaces might include: a core creative framework, a media production project, a product development track, an organizational domain, etc.

Workspaces are:

- Separated by default so professional, creative, and organizational work don't bleed together
- Cross-referenced when topics intersect - a lightweight link between workspaces records that concept A in workspace X was discussed alongside concept B in workspace Y
- Mergeable when domains genuinely converge - a merge zone creates shared visibility between two workspaces while preserving both histories
- Expandable - new workspaces can be created at any time with no limit

Source Ingestion Pipeline

The system accepts source material in any of the following formats:

Source Type	Format	Handling

ChatGPT exports	.md files	Parsed using ## USER / ## ASSISTANT / ## TOOL turn delimiters
Grok exports	.html files (saved from browser)	Parsed using HTML DOM structure (BeautifulSoup)
Claude exports	.md files (manual copy)	Parsed as markdown
Google Docs	.docx (downloaded)	Converted via pandoc, then processed as markdown
Google Sheets	.csv or .xlsx	Parsed as structured data
PDFs	.pdf	Text extraction via pdftotext/pymupdf; OCR fallback for scanned documents
Desktop files	Any text-based format	Direct read
Images	.png, .jpg, etc.	Metadata logged; flagged for manual review

Files that cannot be parsed (corrupted, encrypted, unusual format) are logged and skipped - they never block the pipeline.

Core Innovation - Chat Segmentation Engine

This is the core innovation. Long AI conversations routinely cover multiple topics in a single session. The system splits each conversation into segments - contiguous blocks that discuss a coherent topic.

Each segment captures:

- Topic summary - what was discussed in this block
- Exact source location - file path, start/end line numbers or page numbers
- Date - when the conversation occurred
- Platform - which AI platform it came from
- Workspace assignment - which workspace(s) this segment belongs to
- Concepts detected - which ideas, frameworks, or entities appear in this segment
- Key quotes - the most significant statements, with exact line references
- Cross-references - connections to concepts in other workspaces

- Persona/character detection - when the AI was speaking as a specific character or in a specific voice (relevant for users who develop AI personas)

A single chat file may produce one segment or twenty, depending on how many topics it covers. Each segment is independently searchable and traceable.

Concept Extraction & Registry

After segmentation, the system identifies all recurring concepts across all segments and workspaces. A concept is any idea, framework, entity, project, symbol, or thread that appears across multiple conversations.

Each concept entry includes:

- Name and aliases - all the ways this concept has been referred to
- Category - what type of concept (framework, project, person, philosophical thread, etc.)
- Primary workspace - where it lives
- Summary - synthesized from all sources
- Meaning layers - when a concept carries multiple layers of significance, each layer is documented separately with its own evolution history
- Status summary (see below)
- Source segments - every segment where this concept appears, with provenance
- Cross-workspace references - links to related concepts in other workspaces
- Lifecycle state - active, archived, or forgotten

Status Summaries

Every concept gets a living status assessment that answers:

1. **What is this?** - Current synthesized definition
2. **Where did it come from?** - First appearance, how it evolved over time
3. **Where is it now?** - Most recent state of thinking
4. **What's settled?** - Aspects that are stable and decided
5. **What's open?** - Unresolved questions, contradictions, areas not yet explored

6. **What's connected**? - Related concepts within and across workspaces
7. **Source coverage** - Which platforms have discussed this, how deeply, how recently

These summaries make it possible to quickly assess the maturity and completeness of any concept area without reading every source conversation.

Concept Graph

A relationship map captures connections between concepts:

- Within-workspace relationships - how concepts relate to each other (tension, complement, dependency, evolution, etc.)
- Cross-workspace relationships - how concepts in different workspaces connect (foundation, dependency, parallel, convergence)

Each relationship is documented with the source segments that establish it.

Interactive MWML Features

The output is a single-page web application with the following views:

1. Workspace Switcher

Home screen showing all workspaces as cards with concept count, source count, last updated date, and quick status indicators (e.g., concepts needing attention).

2. Constellation Map

Visual node-and-edge graph of concepts and their relationships within a workspace. **Clicking a node reveals all sources, meaning layers, and status.** Cross-workspace links appear as dotted lines with workspace badges. Node glow/size indicates depth of coverage.

3. Concept Explorer

Filterable, scrollable list of all concepts in a workspace. Filters include: category, platform, status (well-developed / in-progress / thin / needs-attention), lifecycle state, and date range. Each card shows summary, meaning layers, source count, and cross-workspace links. Expandable detail view shows full provenance.

4. Source Timeline

Chronological display of all source material feeding a workspace. Filterable by platform. Each entry shows date, platform, chat/file title, segments within it, and which concepts each segment touches. Click-through to exact source location with surrounding context.

5. Search

Full-text search across all concepts, summaries, meaning layers, source excerpts, and context notes. Results grouped by workspace with provenance links. Toggle: include archived / include forgotten items.

6. Status Dashboard

Overview of a workspace showing: concept count by category and completeness, platform coverage gaps, timeline of concept evolution, open questions across all concepts, recently active vs. dormant concepts, and cross-workspace connection summary.

7. Gap Finder

Highlights: concepts with few sources, topics discussed on one platform but not others, concepts not discussed in 6+ months, open questions flagged across concepts, missing connections (concepts that probably should be linked but aren't).

8. Cross-Reference Map

Meta-view showing all workspaces as nodes with connection lines between them. Line thickness indicates density of cross-references. Click to see specific cross-references.

9. Merge Zones

Shows active convergence points between workspaces - what's connecting, what's shared, what's still separate.

10. Archive View

All archived and forgotten items across all workspaces. Sortable, filterable. Each item shows: reason for archival, date, original workspace. One-click restore.

Concept Management

Drag-and-Drop Organization

Concepts can be dragged between workspaces, into the archive, or into the forget bin. Bulk operations supported (select multiple → move/archive/forget).

Three-State Lifecycle

State	Visibility	Purpose
Active	Visible everywhere	Default state - included in all views, search, and status summaries
Archived	Hidden from default views; visible in search with toggle	Soft-hide for outdated or low-priority items. Fully restorable.
Forgotten	Hidden from everything; visible only with explicit toggle	Deep-hide for truly irrelevant items. Data preserved, never deleted. Fully restorable.

Movement History

When a concept is moved between workspaces, a moved_from history entry is created. **Provenance is never lost.**

Incremental Updates

Adding New Material

New source files can be added at any time without reprocessing the entire corpus:

1. Drop files into the sources directory (or run a command specifying the file/folder)
2. System ingests only the new files
3. Segments new conversations and assigns to workspaces
4. Matches new concepts to existing registry (or creates new entries)
5. Generates cross-references for any cross-workspace mentions
6. Updates status summaries for all affected concepts
7. Rebuilds the MWML with updated data

Console output summarizes what changed: "Added 3 files → 12 segments → 4 new concepts, 8 existing concepts updated, 2 cross-references created."

Same-Chat Updates

When a conversation has been continued since the last export, re-exporting and re-importing the file causes the system to diff from the end of the previously ingested content and process only the new material.

Explain Mode

Generates clear, audience-appropriate explanations of any workspace or concept set. Designed for communicating the work to external audiences.

Audience Calibration

Audience	Output Style
Newcomer	The story. Why it matters. Big picture. No jargon.
Domain-familiar	The framework. How concepts relate to known ideas. What's novel.
Collaborator	The status. What's built, what's in progress, where to plug in.
Technical	The architecture. How pieces connect. Dependencies. Implementation state.

Output

Generated as markdown and/or HTML, saved to an exports directory. Includes a provenance appendix showing which concepts and sources informed each section.

Technical Implementation

- Built with Claude Code - the entire system is constructed through Claude Code sessions that read source files and generate the indexes and output
- Runs locally - all data stored on the user's machine, no cloud hosting required
- Single-file output - the MWML is a self-contained HTML file loading from a single data.json; opens in any browser.
- Current implementation: a Next.js reader app at apps/mwml-reader/. A future build can produce a single-file HTML version per the original design.
- No external dependencies - runs offline after build
- Incremental processing - new material is processed without re-reading the entire corpus
- Cost-managed - work happens in focused sessions (one batch per session); heaviest token usage is during concept extraction; rebuilding the HTML is lightweight

Potential Applications

While built for consolidating creative and strategic work across AI platforms, this architecture could be applied to:

- **Research program management** - tracking the evolution of research threads across papers, conversations, and working documents
- **Product strategy consolidation** - unifying product thinking scattered across Slack, docs, meetings, and AI conversations
- **Organizational knowledge mapping** - discovering what teams know, where knowledge lives, and where gaps exist
- **Multi-stakeholder project alignment** - showing how separate workstreams connect and where they're converging
- **AI-augmented research workflows** - preserving and organizing the outputs of multi-model research processes (e.g., running the same analysis through multiple AI platforms and tracking the synthesis)